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2. $\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$ $s = \sqrt{\frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2}$

$ME = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}}$

then $\bar{x} = 3.0625$
 $s = 0.5086$

for 90% ($\alpha = 0.1$, $\alpha/2 = 0.05$) $df = 7$, $t_{0.05, 7} \approx 1.895$
 for 95% $df = 7$, $t_{0.025, 7} \approx 2.365$
 for 99% $df = 7$, $t_{0.005, 7} \approx 3.499$

$\therefore ME_{90} = 0.0314$
 $ME_{95} = 0.423$
 $ME_{99} = 0.624$

90% (2.7215, 3.4035)
 95% (2.6395, 3.4855)
 99% (~~3.0625~~ 2.4385, 3.6865)

4. normal distribution of σ^2 and μ is given by $(\bar{X} - 2\sigma_{\bar{X}}, \bar{X} - 2\sigma_{\bar{X}})$

$2\sigma_{0.025} \approx 1.96$
 $2 \cdot 1.96 \frac{s}{\sqrt{n}} < 0.010$
 $\therefore 3.92 \cdot \frac{1}{\sqrt{n}} < 0.01$
 $n = 153665$

7. $\bar{x} = 156.85$
 $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1} = 491.7789$
 $\therefore s = 22.14$

$ME = t_{\alpha/2} \cdot \frac{s}{\sqrt{n}} \approx 8.36$
 $CI = \bar{x} \pm ME = (148.49, 165.21)$

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2. 7. $S_x^2 = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})^2$
 $S_y^2 = \frac{1}{m-1} \sum_{i=1}^m (y_i - \bar{y})^2$

$E(S_x^2) = \sigma^2$ $E(S_y^2) = 2\sigma^2$
 $E(\alpha S_x^2 + \beta S_y^2) = \sigma^2 = \alpha E(S_x^2) + \beta E(S_y^2)$

$\alpha \sigma^2 + \beta (2\sigma^2) = \sigma^2$
 $\therefore \alpha + 2\beta = 1$
 $\therefore$ the solution is $\alpha + 2\beta = 1$

$Var(S_x^2) = \frac{2\sigma^4}{n-1}$ $Var(S_y^2) = \frac{4\sigma^4}{m-1}$

$\therefore Var(\alpha S_x^2 + \beta S_y^2) = \alpha^2 \frac{2\sigma^4}{n-1} + \beta^2 \frac{4\sigma^4}{m-1}$
 $\therefore \alpha + 2\beta = 1$
 $\therefore Var(\alpha S_x^2 + \beta S_y^2) = \frac{2\sigma^4}{n-1} - \frac{8\sigma^4}{n-1} + \beta^2 (\frac{8\sigma^4}{n-1} + \frac{4\sigma^4}{m-1})$
 $\therefore dVar(\alpha S_x^2 + \beta S_y^2) = 0$
 $\therefore \beta = \frac{(m-1)(n-1)}{2m+n-3}$ $\alpha = 1 - \frac{2(m-1)(n-1)}{2m+n-3}$

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2. $n = 9$, $\bar{x} = 22$, $\sum_{i=1}^n (x_i - \bar{x})^2 = 72$

a) $H_0: \mu \leq 20$, $H_1: \mu > 20$
 $s = \sqrt{\frac{72}{8}} = 3$
 $t = \frac{\bar{x} - \mu_0}{\frac{s}{\sqrt{n}}} = 2$
 $df = n-1 = 8$
 $t_{0.05, 8} \approx 1.860$ $\therefore t = 2 > 1.860$
 $\therefore$ refuse H_0 , $\Rightarrow \mu > 20$

b) $H_0: \mu = 20$, $H_1: \mu \neq 20$ $t = 2$
 $df = 8$, $t_{0.025, 8} \approx 2.306$
 $|t| = 2 < 2.306$, can't refuse H_0 .
 $\therefore$ in 95% , can't refuse H_0 , $\mu = 20$.

c) in 95% $\bar{x} \pm t_{\alpha/2, df} \cdot \frac{s}{\sqrt{n}}$ is the limit.
 $\alpha = 0.05$, $t_{0.025, 8} \approx 2.306$
 $\therefore$ this is $[19.694, 24.306]$

